

AI-Driven Intelligent Routing and Secure Autonomous Recovery in 6G-Integrated Disaster-Monitoring Sensor Networks

PhD Defence

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Agenda

- 1 Problem and Research Question
- 2 Architecture and Methodology
- 3 Results
- 4 Business Value and Path Forward

The Problem: Networks Fail When Lives Are at Stake



The Disaster Coverage Gap

- Natural disasters cause **\$300B +/year** in global losses
- Sensor networks deployed for monitoring **drain their batteries within hours** under disaster load
- Connectivity collapses exactly when **emergency calls** and alerts are most critical
- Sending field crews into a disaster zone is **dangerous and very costly**



Why This Is a Mobility Operator Problem

- **Coverage liability:** emergency call failures during crises
- **OPEX pressure:** frequent battery swaps and field maintenance
- **ESG gap:** no standardized carbon metric for sensor networks
- **Compliance risk:** AI-managed networks lack certified guardrails

The Core Gap

Existing solutions **cannot simultaneously** guarantee high data delivery, minimize carbon cost, and protect AI-managed networks from compromise — three obligations that 6G mobility operators now face together.

Research Question and Three Objectives

Research Question

Can AI make disaster-monitoring sensor networks fully autonomous — staying alive, carbon-efficient, and trustworthy — without human intervention or expensive infrastructure?

Objective 1

Maximize network lifetime and data delivery

Keep every sensor alive as long as possible. Deliver $\geq 98\%$ of data packets. Run on a battery-powered chip — no relay gateway needed.

Objective 2

Minimize lifecycle carbon footprint

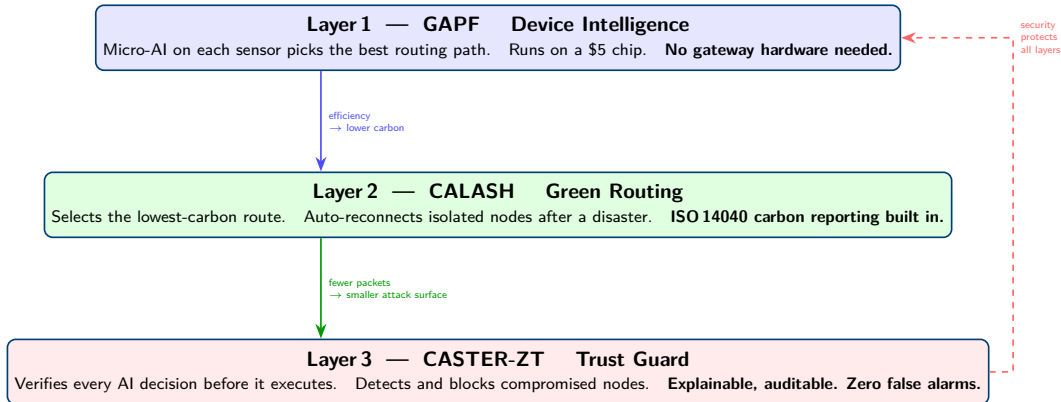
Measure and reduce the carbon cost of every data packet delivered. Self-heal the network after a disaster with no human intervention.

Objective 3

Detect and block AI misbehaviour in real time

Verify every AI decision before it executes. Zero false alarms. Explainable and auditable for regulators.

Solution Architecture: Three Interlocking Layers



The three layers **reinforce each other**. Deployed together on one microcontroller, they cover network lifetime, carbon sustainability, and AI safety — with **no trade-off** between the three objectives.



How We Built and Tested

- Designed three AI tools and stress-tested them in thousands of simulated disaster scenarios (earthquakes, cascading node failures, cyberattacks)
- Each tool independently published in **peer-reviewed IEEE journals**:
GAPF and CALASH in *IEEE TGCN*,
CASTER-ZT in *IEEE TMC*
- Combined: **more than 23,800 test runs** across three independent studies



Three Real-World Data Sources

- **Sensor behaviour**: Intel Research Lab real sensor traces (temperature, humidity, voltage)
- **Carbon intensity**: UK National Grid live carbon API (real grid emissions over time)
- **Disaster geometry**: USGS ShakeMap from the Turkey-Syria earthquake (Mw 7.8, February 2023)

Scope

Results are simulation-based. Hardware validation (Zolertia RE-Mote, Google Coral) is the planned next step.

Result 1 — GAPF: Networks That Survive Disasters

⚡ What GAPF Delivers

- Each sensor runs a micro-AI that learns the best routing path for its battery level and position
- The entire AI fits in **5.8 KB** — smaller than a text message
- No expensive relay gateway is needed: the intelligence lives *on* the sensor

🧰 Business Translation

- **Fewer truck rolls:** sensors last 2–3.5× longer
- **Lower CAPEX:** no relay gateways to buy or install
- **Coverage guarantee:** 98% data delivery even under full disaster load

Metric	Best prior	GAPF
Data delivery	46–67%	≥ 98%
Energy/packet	1.0×	0.3–0.5×
AI model size	~39K	1,473 params
Training memory	3.2–14 GB	951 MB

26× lighter AI model
than the best competing approach,
yet **more reliable** on every metric

Result 2 — CALASH: Networks That Stay Green



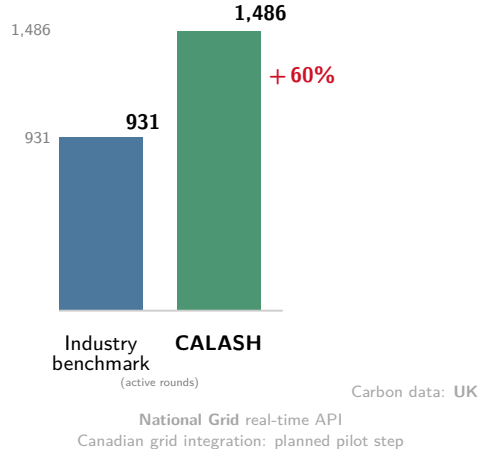
What CALASH Delivers

- Introduces the first **carbon intensity metric** for wireless sensor networks — fully **ISO 14040 compliant**
- Routes data through the path with the **lowest carbon cost**, not just the shortest hop
- Uses 6G links to **reconnect isolated nodes automatically** after a disaster, with no human intervention



Business Translation

- **ESG reporting:** ready-made KPI for CSRD and SEC sustainability requirements
- **60% longer network lifetime** reduces maintenance cycles
- **33% less CO₂ per packet** — auditable for regulators and investors



Result 3 — CASTER-ZT: Networks That Stay Trustworthy



What CASTER-ZT Delivers

- Acts as an **always-on safety guardrail**: every AI decision is verified before it executes
- Five graduated responses (monitor → correct → defer → escalate → block) give **proportional, auditable** reactions
- The safety layer is **fully explainable**: no black box; every decision is traceable



Business Translation

- **0 false alarms**: no service disruption from over-blocking
- **3 ms per decision**: within real-time radio budget (O-RAN limit: 10 ms)
- **Certification path**: auditable design aligned with IEC 62443 and NIST SP 800-207

Metric	Best prior	CASTER-ZT
Threat detection	60–84%	74–98%
False alarms	48.8%	0%
Decision latency	>15 ms	3.07 ms
Recovery score	0.14–0.20	0.733

Tested

over 2,420 scenarios across 100-cell hexagonal networks
Recovery 3.7–5.2× **better** than best competing approach

Business Value for Mobility Operators

Priority	GAPF	CALASH	CASTER-ZT
Reduce OPEX	2–3.5× longer battery life ⇒ fewer field crews	+60% network lifetime ⇒ fewer maintenance cycles	Near-zero manual interventions per security event
Reduce CAPEX	No relay gateways; runs on \$5 sensor chips	Reuses existing 6G and O-RAN infrastructure	Runs on same micro-controller as GAPF
ESG & Sustainability	26× lighter AI ⇒ less compute energy	ISO 14040 carbon KPI ready for CSRD / SEC	Explainable decisions support ESG governance
Regulatory Safety	98% delivery in disaster zones	Standardisable metric (ISO, 3GPP, O-RAN)	IEC 62443 / NIST SP 800-207 path

All three tools fit together on **one microcontroller** (103 KB total). They are **complementary**: efficiency, sustainability, and safety reinforce each other. All algorithms are **open-access with no vendor lock-in** (thesis published by Polytechnique Montréal).

1–2 Years

- Integrate GAPF, CALASH, and CASTER-ZT into a **unified operational stack**
- **Hardware pilot** on real equipment (Zolertia RE-Mote, Google Coral) to confirm lab results in the field
- Extend disaster models to cascading events and wildfires

2–4 Years

- Deploy as **O-RAN apps** (rApps and xApp) on open RAN infrastructure
- Distributed AI training across multiple operator networks, without sharing raw data
- Formal certification of the safety layer (IEC 61508)

4+ Years

- Integrate **satellite and drone** backhaul (LEO, HAPS) for post-disaster coverage in any geography
- Submit LCI carbon metric and zero-trust framework to **ISO, 3GPP, O-RAN Alliance** for standardisation

Transfer domains: Precision agriculture Smart cities Connected health Vehicle and drone networks

Thank You!

Questions?

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Backup Slides

Detailed results — available on request

Full Quantitative Summary — All Three Studies

Metric	Best prior approach	Prior value	Our result	Gain
Data delivery rate	QMIX (RL agent)	46–67%	$\geq$ 98%	+43–111%
Energy per packet	QMIX	1.0×	0.3–0.5×	2–3.5× ↓
AI model parameters	RNN agent	~39K	1,473	26× ↓
Training memory	QMIX	3.2–14 GB	951 MB	3.4–14.5× ↓
Network lifetime	LEACH	931 rounds	1,486 rounds	+60%
Carbon per packet	LEACH	13.42	8.99	–33%
Threat detection	Best 0-FP ref.	0%	74–98%	+74–98 pts
False alarms	Shield-Binary	48.8%	0%	Eliminated
Decision latency	O-RAN budget	10 ms	3.07 ms	3.3× ↓
Criteria covered	Best existing	1 / 12	12 / 12	12× ↑
Total test runs across all three studies				>23,800

The complete AI stack uses **~27,000 parameters** and fits in **103 KB** — on a microcontroller costing a few dollars. This is **38× lighter** than MCUNet, the current reference for AI on embedded chips.

UN Sustainable Development Goals Alignment

SDG	Target	Contribution
9	Resilient infrastructure, innovation	Ultra-light routing AI (1,473 params)
11	Reduce disaster losses (target 11.5)	98% data delivery + automatic self-healing
12	Sustainability reporting (target 12.6)	ISO 14040 carbon KPI for IoT networks
13	Climate resilience (target 13.1)	Carbon-aware routing + post-disaster recovery
16	Transparent institutions (target 16.6)	Explainable safety shield + 0 false alarms

Sendai Framework for Disaster Risk Reduction

Directly addresses **indicators C-2** (communication and warning systems) and **C-6** (early warning infrastructure) of the UN Sendai Framework 2015–2030.

Standardisation Pathway

LCI metric is a candidate for **ISO TC 207/SC 7**. GAPF and CALASH are proposed as rApps for **O-RAN Alliance WG2**. CASTER-ZT maps to **3GPP SA5** AI/ML trust requirements.